

Optimising Wet Chemical Surface Treatment of Biochar

1 Objective

The objective of this study is to develop a systematic understanding of how water and hydrochloric acid (HCl) treatment protocols modify biochar surface chemistry, pore structure, and electrochemical performance. The focus is on identifying an appropriate treatment methodology that enables controlled tuning of surface characteristics while improving initial Coulombic efficiency (ICE), reversible capacity, and voltage profile stability.

Specifically, this study aims to screen:

- Treatment medium (water vs. HCl)
- Treatment severity across a controlled range

The study is designed as a systematic screening of six treatment conditions to identify a processing “sweet spot” that balances surface passivation, defect control, and electrochemical reversibility.

2 Experimental Design Strategy

To enable meaningful trend analysis while remaining experimentally practical, a structured screening design is adopted. The design consists of:

- Two treatment chemistries (water and HCl)
- Six treatment severities spanning mild to aggressive conditions

All samples originate from the same biochar batch (identical precursor and pyrolysis history) to isolate the effects of wet chemical treatment on material properties.

3 Fixed Processing Parameters

The following parameters are held constant across all experiments:

- Biochar precursor and pyrolysis history
- Particle size distribution (post-milling and sieving)
- Solid-to-liquid ratio during treatment
- Treatment temperature (room temperature)
- Post-treatment rinsing protocol (deionised water to neutral pH)
- Post-treatment drying protocol (vacuum drying at 120 °C for 12 h)

4 Variable Parameters

4.1 Treatment Chemistry

Two post-treatment chemistries are investigated:

- Deionised water (baseline physical cleaning and soluble impurity removal)
- Hydrochloric acid (HCl) solution (chemical de-ashing and surface functionalisation)

HCl is selected due to its effectiveness in removing inorganic residues while minimising oxidative damage to the carbon framework.

4.2 Treatment Severity

Treatment severity is systematically varied by adjusting exposure time and/or acid concentration. Three discrete severity levels are defined to span a wide processing window:

- Severity 1: Mild
- Severity 2: Moderate
- Severity 3: High

For water treatments, severity is primarily controlled by washing duration and repetition. For HCl treatments, severity is controlled by acid concentration and exposure time.

5 Treatment Matrix

Table 1: Water and HCl treatment experiment matrix

Run	Treatment Chemistry	Severity Level
1	Water wash	1
2	Water wash	2
3	Water wash	3
4	HCl treatment	1
5	HCl treatment	2
6	HCl treatment	3

6 Characterisation Plan

6.1 Structural and Surface Characterisation

- BET surface area and pore size distribution
- Raman spectroscopy (I_D/I_G ratio)
- XPS
- Elemental analysis and ash content

6.2 Thermal Decomposition and Gas Evolution Analysis

Thermogravimetric analysis coupled with mass spectrometry (TGA–MS) will be used to identify gaseous species evolved during pyrolysis of the biochar and to assess the impact of post-treatment on residual functional groups and thermal stability.

Samples will be heated under an inert atmosphere using a controlled temperature ramp representative of the original pyrolysis conditions. Simultaneous mass spectrometric analysis of the evolved gases will enable identification of key decomposition products, including:

- H_2O (moisture release and dehydration reactions)
- CO and CO_2 (decomposition of oxygen-containing functional groups)
- H_2 (secondary carbon condensation reactions)
- Low-molecular-weight hydrocarbons (e.g. CH_4 , C_2 species)

Comparative TGA–MS analysis across treated and untreated samples will be used to:

- Assess differences in residual oxygen-containing surface groups introduced or removed by water and HCl treatments
- Identify changes in thermal stability and decomposition onset temperatures
- Correlate gas evolution profiles with surface chemistry measured by XPS and Raman spectroscopy

The evolution of CO and CO_2 will be of particular interest as indicators of surface oxygen functionalities that contribute to irreversible capacity loss and initial Coulombic efficiency in electrochemical testing.

6.3 Electrochemical Characterisation

All samples will be evaluated using identical electrode formulations, areal loadings, and formation protocols. Key performance metrics include:

- Initial Coulombic efficiency (ICE)
- Reversible capacity (at $\text{C}/10$)
- Voltage profile shape and plateau capacity fraction
- Early-cycle capacity retention (first 10 cycles)

7 Data Analysis and Method Selection

The following relationships will be analysed:

- ICE as a function of treatment severity for water and HCl
- Reversible capacity vs. BET surface area and surface oxygen content
- Plateau capacity fraction vs. pore structure evolution
- Irreversible capacity loss vs. gas evolution indicators from TGA–MS

Trend-based comparisons between water and HCl treatments will be used to decouple the effects of physical cleaning from chemical surface modification.

8 Decision Criteria

Based on the results:

- If water treatment alone yields comparable improvements in ICE and voltage profile, acid treatment may be unnecessary.
- If HCl treatment shows a clear optimum severity that maximises ICE without excessive capacity loss, this condition will be selected for scale-up.
- If aggressive HCl treatments degrade plateau capacity or increase irreversible loss, treatment severity will be constrained to the identified optimal window.

This screening study will establish a robust, chemically minimal post-treatment methodology for controlling biochar surface chemistry and guiding subsequent electrochemical optimisation.